

ATHARVA AMBADE

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SKILLS

Languages: Java, Python, SQL, Go, C++, JavaScript, TypeScript, Bash/Shell
Frameworks & Backend: Spring Boot, FastAPI, Node.js, RESTful APIs, gRPC, Apache Kafka, Spring Security, JWT
Databases & Data: PostgreSQL, IBM Db2, Redis, MongoDB, Keras, OpenCV, LLMs, GenAI Tools
Cloud, Infra & Tools: AWS (EC2, EKS, S3, RDS), Kubernetes, Docker, Terraform, Jenkins, Git, Linux, Splunk

EDUCATION

California State University, East Bay | Master's Degree in Computer Science May 2026
GPA: 3.8/4.0 | Coursework: Distributed Systems, Advanced Algorithms, Database Systems, Cloud Computing
Savitribai Phule Pune University | Bachelor's Degree in Computer Engineering May 2021
GPA: 3.5/4.0 | Coursework: Data Structures, Operating Systems, Computer Architecture, Computer Networks

EXPERIENCE

Software Engineer Hayward, CA
California State University, East Bay | Student Assistant September 2024 - May 2026

- Developed **Python** automation scripts to synchronize data between internal university systems, reducing repetitive manual data entry for administrative staff by **40%**.
- Implemented **Python** scripts integrating internal university APIs to streamline student records synchronization across administrative systems.
- Built student facing **Reduced Course Load (RCL)** interface on **Salesforce Education Cloud**, digitizing the intake workflow and reducing manual processing time by **60%** for 10,000+ students.
- Assisted in configuring and testing a campus AI chatbot with a basic **RAG** backend to automatically route student support tickets to the correct department.

Software Engineer Pune, India
Accenture January 2022 - June 2024

- Contributed to decomposing a monolithic banking application into independently deployable **microservices** using **Java Spring Boot**, improving system modularity and scaling for 2M+ customers.
- Identified and resolved **IBM Db2** query bottlenecks on banking transaction tables, applying B-tree indexing strategies that decreased P99 latency by **35%** under peak concurrent load.
- Built secure **RESTful APIs** using **Java Spring Boot** with **RBAC** and **JWT**, enforcing rate limiting on financial endpoints handling high volume RPS across microservices.
- Implemented **Apache Kafka** producers and **Spring Boot** consumers to decouple transaction ingestion from legacy settlement modules, utilizing DLQ routing to prevent data loss.
- Designed and implemented **Python** and **Control-M** batch ETL jobs end to end with exactly once execution semantics, processing **15+ TB** of data and reducing failure rates by **40%**.

PROJECTS

Distributed MCP Server | Python, AsyncIO, Redis, Llama 3.2, MCP Protocol | [Link](#)

- Architected a concurrent **MCP** server with async I/O and connection pooling, solving distributed race conditions in API token management via the **Redlock** algorithm and **Redis**.
- Built an async **Python** MCP server with non blocking I/O, integrating **Llama 3.2** with context window management to reduce token usage by **60%** while maintaining **95%** command accuracy.

Plagiarism Detector ML Model | Python, Keras, CNN, OpenCV, dlib | [ICINC 2021 Publication](#)

- Developed a published machine learning proctoring model, training a **MobileNetV2 Convolutional Neural Network (CNN)** via **Keras** to classify real time facial orientation with **99%** accuracy.
- Designed a video preprocessing pipeline using **OpenCV** and **dlib** to extract facial features, optimizing batch sizes and learning rates for low latency inference.

Smart Calorie Counter | Flask, PostgreSQL, Redis, AWS EKS, Kubernetes, Docker | [Link](#)

- Built a **Flask** REST API with **PostgreSQL** storage and designed a **Redis** caching strategy that cut external API calls by **70%** and reduced response times to under **300ms**.
- Deployed full stack application to **AWS EKS** using **Docker** and **Kubernetes** manifests with horizontal pod autoscaling and ConfigMap based environment configuration.